In[ ]:=
```
n1 = 7;
g1 = 3;
n2 = 7;
g2 = 1;
n3 = 7;
g3 = 1;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.400144239328412;
β12 = 0.20902489540234132;
β22bar = 0.81049344303531;

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.5294117647058824; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -0.9523141799457909;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in the middle region. This is obtained by solving two
  consistency conditions along with other constant γ12bar. *)
Q2 = -1.990947646406977;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
 Q2 f3^(3/2)/Sqrt[x] in the inner region. This is obtained by solving three
    consistency conditions along with other constants β13 & β23. *)
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
R = 3 a/(8 * alfas) * (3 π η/2)^(2/3);(*Page number 54, it is inverse in the copy though *)
c = 12/(5 π) 4/3 g^2 4/3 alfas/a (3 π η/2)^(1/3);
(*Multiplication factor in the expression of kinetic energy *)
d = 4/3 g^2 4/3 alfas/a (3 π η/2)^(1/3) 18/(25 π) η/(η - 1/5);  (* 18g^2/(5(5η -1 )).16 π^2.(3a/8alfas). R^2/(6π^2)^2 ,
The value of R is from page number 54 which is written inverse here,
and (3a/8alfas) is inverse in copy, BA, 13 September *)
(*Multiplication factor in the expression of Potential energy *)
```

(*Let us define the three functions for three regions *)

```
f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
  Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
   (*Appendix from Kinetic energy*) /.
   First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)]/x^(1/2)), y3'[x2] == β22bar * s2 ,
      y3[x2] == β22bar * u2 (* It shows how the function f3
       (x3) changes on other regions*)}, y3, {x, x2, x3}];
(* s2 is the first derivative of function f_2(x) *)
(*u2 is the function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x]  in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
   (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)]/x^(1/2)),
     y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
     (* It means at the boundary x3 the function df1 changes to other function *)
     y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
     (*It means at the boundary x3 the function f1 changes to other function*)
    }, (* It means at the boundary x3 the function f1 changes to other function *)
    y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
```

```
Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
    condition of d/dx f1(x)|_x2 = γ12bar d/dx f2(x)|_x2 and f1(x3) = γ12 f2(x3)  *)

f2[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^(3/2)]/x^(1/2)), y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3/2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2/(3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x3 inner region. On this region,
f1(x)=β13 f3(x) and f2(x)= β23 f3(x).  *)

(*Start the program*)

m2 = 41;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
```

```
m2B = m2;
Energy = 10^9;
Energy = 10^9;
Energy2plus = 10^9;
H = 10^9 + 1;
SN = 0;
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)] / x^(1/2)),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];

  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2,  u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2,  u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];
```

```
x1 = x3 + 0.1;
FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
solution1 = NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)]) / x^(1/2),
    y1'[x3] == γ12bar * s3 , y1[x3] == γ12bar * u3 }, y1, {x, x3, x1}];
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (* u1 is the function f_1(x) *)
s1 = t6[[1]]; (* s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)

normmx2 = x2 s2 - u2 ;
normx2 = (n2 Q2) / (3 η) ;
(*normx2barr and normx2bar should be equal *)
normmx3 = x3 s3 - u3 ;
normx3 = (n3 Q3) / (3 η) + β22bar (n2 Q2) / (3 η) - (β22bar)^(3/2) (n2 Q3) / (3 η);
(*normmx2 and normx2 should be equal *)
normmx1 = x1 s1 - u1 ;
normx1 = (n1 Q1) / (3 η) + γ12bar (n3 Q3) / (3 η) - (γ12bar)^(3/2) (n3 Q1) / (3 η) +
  ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) (n2 Q1) / (3 η) - γ12bar (β22bar)^(3/2) (n2 Q3) / (3 η) + γ12bar β22bar (n2 Q2) / (3 η) ;
(*normmx1 and normx1 should be equal *)

h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
```

```
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
h6 = h3[[1]] ;
```

$$AB = \frac{4}{7}\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{5\,n2}{21\,\eta}\,s2\,;$$

$$CD = \frac{4}{7}\sqrt{x3}\,(u3)^{\frac{5}{2}} - \frac{4}{7}(\beta 22bar)^{\frac{5}{2}}\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{5}{7\,Q3}\,s3\left(\frac{n3\,Q3}{3\,\eta} + \beta 22bar\,\frac{n2\,Q2}{3\,\eta} - (\beta 22bar)^{\frac{3}{2}}\,\frac{n2\,Q3}{3\,\eta}\right) + \frac{5}{7\,Q3}(\beta 22bar)^2\,\frac{n2\,Q2}{3\,\eta}\,s2\,;$$

$$CCD = \frac{4}{7}\sqrt{x3}\,(u3)^{\frac{5}{2}} - \frac{4}{7}\sqrt{x2}\,(\beta 22bar)^{\frac{5}{2}}(u2)^{\frac{5}{2}} - \frac{5}{7\,Q3}\,s3\,normmx3 + \frac{5}{7\,Q3}(\beta 22bar)^2\,normmx2\,s2\,;$$

$$EF = \frac{4}{7}\sqrt{x1}\,(u1)^{\frac{5}{2}} - \frac{4}{7}(\gamma 12bar)^{\frac{5}{2}}\sqrt{x3}\,(u3)^{\frac{5}{2}} - \frac{5}{7\,Q1}\,s1\left(\frac{n1\,Q1}{3\,\eta} + \gamma 12bar\,\frac{n3\,Q3}{3\,\eta} - (\gamma 12bar)^{\frac{3}{2}}\,\frac{n3\,Q1}{3\,\eta} + \frac{n2\,Q1}{3\,\eta}\left((\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - (\beta 12)^{\frac{3}{2}}\right) - (\gamma 12bar)(\beta 22bar)^{\frac{3}{2}}\,\frac{n2\,Q3}{3\,\eta} + (\gamma 12bar)(\beta 22bar)\,\frac{n2\,Q2}{3\,\eta}\right) + \frac{5}{7\,Q1}(\gamma 12bar)^2\,s3\left(\frac{n3\,Q3}{3\,\eta} + \beta 22bar\,\frac{n2\,Q2}{3\,\eta} - (\beta 22bar)^{\frac{3}{2}}\,\frac{n2\,Q3}{3\,\eta}\right);$$

```
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)
```

$$K222 = \frac{n2}{3\,\eta\,Q2}\,s2 - \frac{1}{Q2}\,h4\,;$$

$$K22 = \frac{4\,n2}{7\,\eta\,Q2}\,s2 - \frac{4}{7\,Q2}\sqrt{x2}\,(u2)^{\frac{5}{2}};$$

```
(*K2bar2barr and K2bar2bar should be equal *)
```

$$K32 = -\frac{4}{7\,Q2}\,\beta22bar^{1.5}\sqrt{x2}\,(u2)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2} - \frac{\beta22bar}{Q3} + \frac{5*\beta22bar^{1.5}}{7\,Q2}\right)s2 + \frac{n2}{3\,\eta\,Q3}\,s3\,;$$

$$K332 = \frac{n2\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}\,s2 - \frac{n2\,\beta22bar}{3\,\eta\,Q3}\,s2 + \frac{n2}{3\,\eta\,Q3}\,s3 - \frac{(\beta22bar)^{\frac{3}{2}}}{Q2}\,h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21 = -\frac{4}{7\,Q2}\,\beta12^{1.5}\sqrt{x2}\,(u2)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\,\beta12^{1.5} + \frac{\beta12^{1.5}}{Q2} - \frac{\gamma12bar^{1.5}*\beta22bar}{Q3}\right)s2 + \frac{n2}{3\,\eta}\left(\frac{\gamma12bar^{1.5}}{Q3} - \frac{\gamma12bar}{Q1}\right)s3 + \frac{n2}{3\,\eta\,Q1}\,s1\,;$$

$$K221 = \frac{n2}{3\,\eta\,Q2}\,(\beta12)^{\frac{3}{2}}\,s2 + \frac{n2\,(\gamma12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}\,s3 - \frac{n2}{3\,\eta\,Q3}\,(\gamma12bar)^{\frac{3}{2}}\,\beta22bar\,s2 + \frac{n2}{3\,\eta\,Q1}\,s1 - \frac{n2}{3\,\eta\,Q1}\,\gamma12bar\,s3 - \frac{(\beta12)^{\frac{3}{2}}}{Q2}\,h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31 = \frac{4}{7}\,\beta22bar^{1.5}\left(\frac{\gamma12bar^{1.5}*\beta22bar}{Q3} - \frac{\beta12^{1.5}}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} - \frac{4}{7\,Q3}\,\gamma12bar^{1.5}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\,\beta22bar\left(\frac{5}{7\,Q2}\,\beta22bar^{\frac{1}{2}}*\beta12^{1.5} - \frac{\gamma12bar^{1.5}*\beta22bar^{1.5}}{Q3} - \frac{5\,Q2}{7\,Q3^2}\,\gamma12bar^{1.5}*\beta22bar + \frac{Q2}{Q3^2}\,\gamma12bar^{1.5}*\beta22bar + \frac{1}{Q2}\,\beta12^{1.5}*\beta22bar^{\frac{1}{2}} - \frac{\beta12^{1.5}}{Q3}\right)s2 + \frac{1}{3\,\eta\,Q3}\left(n3*\gamma12bar^{1.5} - \frac{n3\,Q3}{Q1}\,\gamma12bar + n2*\beta12^{1.5} - \frac{n2\,Q2}{Q3}\,\gamma12bar^{1.5}*\beta22bar +\right.$$

$$\frac{5}{7}\gamma 12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta 22bar-n2*\beta 22bar^{1.5}\right)\Big)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122=\frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3-\frac{\beta 22bar}{Q3}s2\right]-\frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4-\frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$-\frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3-\frac{\beta 22bar}{Q3}s2\right]+\frac{n3}{3\,\eta\,Q1}s1-\frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+(\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2+\frac{1}{Q3}s3-\frac{\beta 22bar}{Q3}s2\right]+$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3-\frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\;;\;*)$$

$$K331=\frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3-\frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2-\frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4-\frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5-$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3+\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2+\frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2}-\frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2+\frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3}-\frac{1}{Q1}\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

(*K12 and K122 should be equal*)

$$K33=\frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3}-\frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}}-$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2}+\frac{2}{7}\frac{Q2}{Q3^2}-\frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2+$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3}+\beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}}-\frac{2}{7}\frac{Q2}{Q3}-\frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\;;$$

$$K333=\frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}-\frac{\beta 22bar}{Q3}\right)s3-\frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}-\frac{\beta 22bar}{Q3}\right)s2-$$

$$\frac{(\beta 22bar)^3}{Q2}\,h4 - \frac{1}{Q3}\,h5 + (\beta 22bar)^3\,\frac{n2}{3\,\eta\,Q2}\,s2 - (\beta 22bar)^{\frac{5}{2}}\,\frac{n2}{3\,\eta\,Q3}\,s2 + \frac{n3}{3\,\eta\,Q3}\,s3\,;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta 22bar^{\frac{5}{2}} * \gamma 12bar^3}{Q3} - \frac{\beta 12^3}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma 12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma 12bar^3}{Q3}\right)\sqrt{x3}\,(u3)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q1}\sqrt{x1}\,(u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta 12^3}{Q2} - \frac{\beta 12^{\frac{3}{2}} * \gamma 12bar^{\frac{3}{2}} * \beta 22bar}{Q3} + \frac{Q2}{Q3^2}\,\gamma 12bar^3 * \right.\right.$$

$$\left.\beta 22bar^2 - \frac{\beta 12^{1.5} * \gamma 12bar^{1.5} * \beta 22bar}{Q3}\right) +$$

$$\left.\frac{5\,n2}{21\,\eta}\left(\frac{\beta 12^3}{Q2} - \frac{Q2}{Q3^2}\,\beta 22bar^2 * \gamma 12bar^3\right)\right)s2 +$$

$$\left(\frac{n2}{3\,\eta}\left(\frac{\beta 12^{1.5} * \gamma 12bar^{1.5}}{Q3} - \frac{\beta 12^{1.5} * \gamma 12bar}{Q1} - \frac{\beta 22bar^{1.5} * \gamma 12bar^3}{Q3} + \right.\right.$$

$$\frac{\beta 22bar^{1.5} * \gamma 12bar^{\frac{5}{2}}}{Q1} + \frac{\beta 12^{1.5} * \gamma 12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\,\gamma 12bar^3 * \beta 22bar -$$

$$\frac{\beta 12^{1.5} * \gamma 12bar}{Q1} + \frac{\gamma 12bar^{\frac{5}{2}} * \beta 22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\,\gamma 12bar^2 * \beta 22bar -$$

$$\left.\left.\frac{Q3}{Q1^2}\,\beta 22bar^{1.5} * \gamma 12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma 12bar^3}{Q3} - \frac{2 * \gamma 12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\,\gamma 12bar^2\right) +\right.$$

$$\frac{5}{21\,\eta}\left(\frac{n3}{Q3}\,\gamma 12bar^3 + \frac{n2\;Q2}{Q3^2}\,\gamma 12bar^3 * \beta 22bar - \frac{n2}{Q3}\,\beta 22bar^{1.5} * \gamma 12bar^3 -\right.$$

$$\left.\left.\left.\frac{n3\ Q3}{Q1^2}\,\gamma 12bar^2 - \frac{n2\ Q2}{Q1^2}\,\gamma 12bar^2 * \beta 22bar + \frac{n2\ Q3}{Q1^2}\,\gamma 12bar^2 * \beta 22bar^{1.5}\right)\right) *$$

$$s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\,\gamma 12bar + \frac{n3}{3\,\eta\,Q1}\,\gamma 12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}\right.$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\,\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\,\beta 22bar^{1.5} * \gamma 12bar\right) +$$

$$\frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\,\gamma 12bar - \frac{n3}{Q1}\,\gamma 12bar^{1.5} + \frac{n2}{Q1}\,\gamma 12bar^{1.5} * \beta 22bar^{1.5} -\right.$$

$$\left.\left.\frac{n2}{Q1}\,\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\,\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\,\gamma 12bar\ \beta 22bar\right)\right) * s1\ ;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}\,h4 - \frac{(\gamma 12bar)^3}{Q3}\,h5 - \frac{1}{Q1}\,h6 + \frac{n2}{3\,\eta\,Q3}\,(\beta 12)^{\frac{3}{2}}\,(\gamma 12bar)^{\frac{3}{2}}\,s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}\,(\gamma 12bar)^3$$

$$\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}\,(\beta 12)^{\frac{3}{2}}\,(\gamma 12bar)^{\frac{3}{2}}\,\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}\,(\gamma 12bar)^3\,(\beta 22bar)^2\,s2$$

$$+ \left(\frac{n2}{3\,\eta\,Q1}\,(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}\,(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\,\gamma 12bar - \frac{n2}{3\,\eta\,Q1}\,(\gamma 12bar)^{\frac{3}{2}}\,(\beta 22bar)^{\frac{3}{2}} -\right.$$

$$\left.\frac{n2\,Q2}{3\,\eta\,Q1^2}\,\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}\,(\beta 22bar)^{\frac{3}{2}}\,\gamma 12bar\right) s1 -$$

$$\left(\frac{n2}{3\,\eta\,Q1}\,(\beta 12)^{\frac{3}{2}}\,\gamma 12bar + \frac{n3}{3\,\eta\,Q1}\,(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\,(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}\,(\gamma 12bar)^{\frac{5}{2}}\right.$$

$$\left.(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\,(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}\,(\beta 22bar)^{\frac{3}{2}}\,(\gamma 12bar)^2\right) s3 +$$

$$\frac{n2}{3\,\eta}\,\frac{(\beta 12)^3}{Q2}\,s2 - \frac{n2}{3\,\eta\,Q3}\,(\beta 12)^{\frac{3}{2}}\,(\gamma 12bar)^{\frac{3}{2}}\,\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}\,(\beta 12)^{\frac{3}{2}}\,(\gamma 12bar)^{\frac{3}{2}}\,s3 -$$

$$\frac{n2}{3\,\eta\,Q1}\,(\beta 12)^{\frac{3}{2}}\,\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}\,(\gamma 12bar)^3\,s3 - \frac{n2}{3\,\eta\,Q3}\,(\beta 22bar)^{\frac{3}{2}}\,(\gamma 12bar)^3\,s3 -$$

$$\frac{n3}{3\,\eta\,Q1}\,(\gamma 12bar)^{\frac{5}{2}}\,s3 + \frac{n2}{3\,\eta\,Q1}\,(\beta 22bar)^{\frac{3}{2}}\,(\gamma 12bar)^{\frac{5}{2}}\,s3 + \frac{n1}{3\,\eta\,Q1}\,s1\ ;$$

```
(*K11 and K111 should be equal *)
```

$$T1 = c\left(\frac{4}{7}\left(\text{alfa1}*\text{g1}\left(\beta 12^{\frac{5}{2}} - \gamma 12\text{bar}^{\frac{5}{2}}*\beta 22\text{bar}^{\frac{5}{2}}\right) + \text{alfa2}*\text{g2}\right)\sqrt{\text{x2}}\ (\text{u2})^{\frac{5}{2}} + \right.$$

$$\frac{4}{7}\text{alfa2}*\text{g3}\sqrt{\text{x3}}\ (\text{u3})^{\frac{5}{2}} + \frac{4}{7}\text{alfa1}*\text{g1}\sqrt{\text{x1}}\ (\text{u1})^{\frac{5}{2}} +$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^2\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}} + \text{alfa2}*\text{g3}\right) - \right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}} - \text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}} - \text{alfa2}*\text{g2}\right)\text{s2} +$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^2*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3} + \beta 22\text{bar}*\text{n2}*\text{Q2} - \beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right) - \right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}} + \text{alfa2}*\text{g3}\right)\left(\text{n3} + \beta 22\text{bar}\frac{\text{n2}*\text{Q2}}{\text{Q3}} - \beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3} -$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1} + \gamma 12\text{bar}\frac{\text{n3}\,\text{Q3}}{\text{Q1}} + \gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2} - \text{n3}\right) - \right.$$

$$\left.\beta 12^{1.5}*\text{n2} - \gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\frac{\text{n2}\,\text{Q3}}{\text{Q1}} + \gamma 12\text{bar}\,\beta 22\text{bar}\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$V1 = -\text{d}\left(\frac{(\text{g1}\,\text{n1} - 1)}{\text{n1}}\text{g1}\,\text{K11} + \frac{(\text{g3}\,\text{n3} - 1)}{\text{n3}}\text{g3}\,\text{K33} + \right.$$

$$\left.\frac{(\text{g2}\,\text{n2} - 1)}{\text{n2}}\text{g2}\,\text{K22} + 2\ \text{g1}\,\text{g3}\,\text{K31} + \frac{4}{3}\text{g2}\,\text{g1}\,\text{K21} + \frac{4}{3}\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
```

```
    Print["normx3 (computationally)=  ", normmx3];
    Print["normx3 (by hand)=  ", normx3];
    Print["normx1 (computationally)=  ", normmx1];
    Print["normx1 (by hand)=  ", normx1];
    Print["0 to x2 (h4 computationally)=  ", h4];
    Print["0 to x2 (h4 by hand)=  ", AB];
    Print["x2 to x3 (h5 computationally)=  ", h5];
    Print["x2 to x3 (h5 by hand)=  ", CD];
    Print["x2 to x3 (h5 by handddddd)=  ", CCD];
    Print["x3 to x1 (h6 computationally)=  ", h6];
    Print["x3 to x1 (h6 by hand)=  ", EF];
    Print["K22  (computationally) = ", K222];
    Print["K22  (by hand) = ", K22];
    Print["K32  (computationally) = ", K332];
    Print["K32  (by hand) = ", K32];
    Print["K21  (computationally) = ", K221];
    Print["K21  (by hand) =", K21];
    Print["K31  (computationally) = ", K331];
    Print["K31  (by hand) = ", K31];
    Print["K33  (computationally) = ", K333];
    Print["K33  (by hand) = ", K33];
    Print["K11  (computationally) = ", K111];
    Print["K11  (by hand) = ", K11];
    Print["x2= ", x2];
    Print["x3= ", x3];
    Print["x1=  ", x1];
     Print["T by hand=", T1];
    Print["V by hand=", V1];
    Print["m2=", m2];
    Print["R=  ", R];
Print["Energywithoutmass = ", Energywithoutmass];
    Print["Energy=  ", Energy];
    If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
      Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
       H = 10^9 + 1;
       m2 = m2B;
       If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
   If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
   If[n == 3, Energy6bare = HX;
    Print["x1=", x1];
    Print["x3=", x3];
    Print["x2=", x2];
```

```
    Print["Kinetic Energy=", T1];
    Print["Potential Energy=", V1];
    Print["Volume Energy=", Volumeenergy];
   ]]];

(*Code for Plotting function *)
Plot1 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar* y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12* y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar* y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

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FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

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FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.0364300575
normx2 (computationally)=  -0.663649121
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099941
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.239283
0 to x2 (h4 by hand)=  13.2392825
x2 to x3 (h5 computationally)=  2.82030189
x2 to x3 (h5 by hand)=  2.82030396
x2 to x3 (h5 by handdddd)=  2.82030315
x3 to x1 (h6 computationally)=  3.35295053
x3 to x1 (h6 by hand)=  3.35295143
K22  (computationally) = 0.844110545
K22  (by hand) = 0.844110273
K32  (computationally) = 0.794753921
K32  (by hand) = 0.794753723
K21  (computationally) = 0.513128838
K21  (by hand) =0.513128812
K31  (computationally) = 0.508191455
K31  (by hand) = 0.508191984
K33  (computationally) = 0.757868383
K33  (by hand) = 0.757870407
K11  (computationally) = 0.443876672
K11  (by hand) = 0.4438785
x2= 0.158694202
x3= 0.176949554
x1=  0.294646997
T by hand=906.258066
V by hand=-193.326791
m2=41
R=  0.0364300575
Energywithoutmass = 1407.35304
Energy=  29575.353
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.0364300575
normx2 (computationally)=  -0.663649121
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099935
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.2425645
0 to x2 (h4 by hand)=  13.242564
x2 to x3 (h5 computationally)=  2.82101122
x2 to x3 (h5 by hand)=  2.82101329
x2 to x3 (h5 by handdddd)=  2.82101247
x3 to x1 (h6 computationally)=  3.35381176
x3 to x1 (h6 by hand)=  3.35381257
K22  (computationally) = 0.844214671
K22  (by hand) = 0.844214398
K32  (computationally) = 0.794852111
K32  (by hand) = 0.794851911
K21  (computationally) = 0.513193262
K21  (by hand) =0.513193236
K31  (computationally) = 0.508255287
K31  (by hand) = 0.508255818
K33  (computationally) = 0.757962146
K33  (by hand) = 0.757964174
K11  (computationally) = 0.443932573
K11  (by hand) = 0.443934249
x2= 0.158674534
x3= 0.176927535
x1=  0.294609399
T by hand=906.487959
V by hand=-193.350986
m2=41.01
R=  0.0364300575
```

```
Energywithoutmass = 1407.29294
Energy=  29575.2929
R=  0.0364300575
normx2 (computationally)=  -0.663649112
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099946
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.2720984
0 to x2 (h4 by hand)=  13.2720977
x2 to x3 (h5 computationally)=  2.82739539
x2 to x3 (h5 by hand)=  2.82739761
x2 to x3 (h5 by handdddd)=  2.82739665
x3 to x1 (h6 computationally)=  3.36156442
x3 to x1 (h6 by hand)=  3.3615654
K22  (computationally) = 0.845151171
K22  (by hand) = 0.845150843
K32  (computationally) = 0.795735245
K32  (by hand) = 0.795735006
K21  (computationally) = 0.513772853
K21  (by hand) =0.513772821
K31  (computationally) = 0.508829596
K31  (by hand) = 0.508830164
K33  (computationally) = 0.758805623
K33  (by hand) = 0.758807783
K11  (computationally) = 0.444437994
K11  (by hand) = 0.44443998
x2= 0.158497843
x3= 0.176729728
x1=  0.294271696
T by hand=908.557421
V by hand=-193.569262
m2=41.1
R=  0.0364300575
Energywithoutmass = 1406.75978
```

```
Energy=  29574.7598
R=  0.0364300575
normx2 (computationally)=  -0.66364917
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099933
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.3049156
0 to x2 (h4 by hand)=  13.3049157
x2 to x3 (h5 computationally)=  2.83448736
x2 to x3 (h5 by hand)=  2.83448859
x2 to x3 (h5 by handdddd)=  2.83448862
x3 to x1 (h6 computationally)=  3.37017841
x3 to x1 (h6 by hand)=  3.37017919
K22  (computationally) = 0.846191259
K22  (by hand) = 0.846191296
K32  (computationally) = 0.796715795
K32  (by hand) = 0.796715822
K21  (computationally) = 0.514416089
K21  (by hand) =0.514416093
K31  (computationally) = 0.509466518
K31  (by hand) = 0.509466849
K33  (computationally) = 0.75974026
K33  (by hand) = 0.759741576
K11  (computationally) = 0.444997737
K11  (by hand) = 0.444999298
x2= 0.158302213
x3= 0.176510708
x1=  0.293897794
T by hand=910.856729
V by hand=-193.811149
m2=41.2
R=  0.0364300575
Energywithoutmass = 1406.18365
Energy=  29574.1837
```

```
R=  0.0364300575
normx2 (computationally)=  -0.66364908
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099933
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.3377306
0 to x2 (h4 by hand)=  13.3377296
x2 to x3 (h5 computationally)=  2.84158356
x2 to x3 (h5 by hand)=  2.84158636
x2 to x3 (h5 by handdddd)=  2.84158483
x3 to x1 (h6 computationally)=  3.3787932
x3 to x1 (h6 by hand)=  3.37879399
K22  (computationally) = 0.847228463
K22  (by hand) = 0.847227928
K32  (computationally) = 0.797694221
K32  (by hand) = 0.797693831
K21  (computationally) = 0.515058468
K21  (by hand) =0.515058417
K31  (computationally) = 0.510103588
K31  (by hand) = 0.510104296
K33  (computationally) = 0.76067702
K33  (by hand) = 0.760679679
K11  (computationally) = 0.445557594
K11  (by hand) = 0.445559259
x2= 0.158107289
x3= 0.176292509
x1=  0.293525307
T by hand=913.156306
V by hand=-194.053086
m2=41.3
R=  0.0364300575
Energywithoutmass = 1405.62436
Energy=  29573.6244
R=  0.0364300575
```

```
normx2 (computationally)=  -0.66364917
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099939
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.3705498
0 to x2 (h4 by hand)=  13.3705499
x2 to x3 (h5 computationally)=  2.84867515
x2 to x3 (h5 by hand)=  2.84867639
x2 to x3 (h5 by handdddd)=  2.84867642
x3 to x1 (h6 computationally)=  3.38740866
x3 to x1 (h6 by hand)=  3.38740954
K22  (computationally) = 0.848266395
K22  (by hand) = 0.848266431
K32  (computationally) = 0.798672605
K32  (by hand) = 0.798672631
K21  (computationally) = 0.51570021
K21  (by hand) =0.515700213
K31  (computationally) = 0.510738808
K31  (by hand) = 0.510739141
K33  (computationally) = 0.761608666
K33  (by hand) = 0.761609994
K11  (computationally) = 0.446116764
K11  (by hand) = 0.446118503
x2= 0.157913095
x3= 0.176075096
x1=  0.293154211
T by hand=915.455927
V by hand=-194.294548
m2=41.4
R=  0.0364300575
Energywithoutmass = 1405.08196
Energy=  29573.082
R=  0.0364300575
normx2 (computationally)=  -0.663649166
```

```
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099946
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.4033678
0 to x2 (h4 by hand)=  13.4033678
x2 to x3 (h5 computationally)=  2.85576954
x2 to x3 (h5 by hand)=  2.85577087
x2 to x3 (h5 by handdddd)=  2.85577082
x3 to x1 (h6 computationally)=  3.39602465
x3 to x1 (h6 by hand)=  3.39602564
K22  (computationally) = 0.849302035
K22  (by hand) = 0.849302044
K32  (computationally) = 0.799649209
K32  (by hand) = 0.799649216
K21  (computationally) = 0.516341111
K21  (by hand) =0.516341111
K31  (computationally) = 0.511373833
K31  (by hand) = 0.511374186
K33  (computationally) = 0.76254127
K33  (by hand) = 0.762542668
K11  (computationally) = 0.446675455
K11  (by hand) = 0.446677409
x2= 0.157719603
x3= 0.175858488
x1=  0.292784504
T by hand=917.755712
V by hand=-194.535862
m2=41.5
R=  0.0364300575
Energywithoutmass = 1404.55615
Energy=  29572.5562
R=  0.0364300575
normx2 (computationally)=  -0.663649168
normx2 (by hand)=  -0.663649215
```

```
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.39709994
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.4361865
0 to x2 (h4 by hand)=  13.4361865
x2 to x3 (h5 computationally)=  2.862864
x2 to x3 (h5 by hand)=  2.86286529
x2 to x3 (h5 by handdddd)=  2.86286529
x3 to x1 (h6 computationally)=  3.40464085
x3 to x1 (h6 by hand)=  3.40464175
K22  (computationally) = 0.850336502
K22  (by hand) = 0.850336527
K32  (computationally) = 0.80062468
K32  (by hand) = 0.800624698
K21  (computationally) = 0.516981214
K21  (by hand) =0.516981216
K31  (computationally) = 0.512008022
K31  (by hand) = 0.512008366
K33  (computationally) = 0.7634726
K33  (by hand) = 0.763473967
K11  (computationally) = 0.447232896
K11  (by hand) = 0.447234696
x2= 0.157526818
x3= 0.175642672
x1=  0.292416167
T by hand=920.055515
V by hand=-194.776735
m2=41.6
R=  0.0364300575
Energywithoutmass = 1404.04685
Energy=  29572.0469
R=  0.0364300575
normx2 (computationally)=  -0.663649168
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
```

```
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099941
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.4690057
0 to x2 (h4 by hand)=  13.4690058
x2 to x3 (h5 computationally)=  2.86995878
x2 to x3 (h5 by hand)=  2.86996007
x2 to x3 (h5 by handdddd)=  2.86996007
x3 to x1 (h6 computationally)=  3.41325769
x3 to x1 (h6 by hand)=  3.41325861
K22  (computationally) = 0.851369698
K22  (by hand) = 0.851369723
K32  (computationally) = 0.801598962
K32  (by hand) = 0.80159898
K21  (computationally) = 0.517620559
K21  (by hand) =0.517620561
K31  (computationally) = 0.512641476
K31  (by hand) = 0.512641823
K33  (computationally) = 0.764402868
K33  (by hand) = 0.764404242
K11  (computationally) = 0.44778994
K11  (by hand) = 0.447791762
x2= 0.157334734
x3= 0.175427644
x1=  0.292049202
T by hand=922.355462
V by hand=-195.017389
m2=41.7
R=  0.0364300575
Energywithoutmass = 1403.55391
Energy=  29571.5539
R=  0.0364300575
normx2 (computationally)=  -0.66364917
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
```

```
normx1 (computationally)=  -0.397099932
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.5018257
0 to x2 (h4 by hand)=  13.5018257
x2 to x3 (h5 computationally)=  2.87705376
x2 to x3 (h5 by hand)=  2.87705503
x2 to x3 (h5 by handdddd)=  2.87705506
x3 to x1 (h6 computationally)=  3.42187483
x3 to x1 (h6 by hand)=  3.42187563
K22  (computationally) = 0.852401675
K22  (by hand) = 0.852401712
K32  (computationally) = 0.802572088
K32  (by hand) = 0.802572115
K21  (computationally) = 0.518259117
K21  (by hand) =0.51825912
K31  (computationally) = 0.513274139
K31  (by hand) = 0.51327448
K33  (computationally) = 0.765331979
K33  (by hand) = 0.765333333
K11  (computationally) = 0.448345925
K11  (by hand) = 0.448347518
x2= 0.157143347
x3= 0.175213399
x1=  0.291683592
T by hand=924.655455
V by hand=-195.257654
m2=41.8
R=  0.0364300575
Energywithoutmass = 1403.07719
Energy=  29571.0772
R=  0.0364300575
normx2 (computationally)=  -0.663649134
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
```

```
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.5346454
0 to x2 (h4 by hand)=  13.5346451
x2 to x3 (h5 computationally)=  2.88415001
x2 to x3 (h5 by hand)=  2.88415192
x2 to x3 (h5 by handdddd)=  2.88415132
x3 to x1 (h6 computationally)=  3.43049276
x3 to x1 (h6 by hand)=  3.43049362
K22  (computationally) = 0.853432
K22  (by hand) = 0.853431807
K32  (computationally) = 0.803543808
K32  (by hand) = 0.803543668
K21  (computationally) = 0.51889691
K21  (by hand) =0.518896891
K31  (computationally) = 0.513906302
K31  (by hand) = 0.513906795
K33  (computationally) = 0.76626082
K33  (by hand) = 0.766262718
K11  (computationally) = 0.448901831
K11  (by hand) = 0.448903577
x2= 0.156952651
x3= 0.174999936
x1=  0.291319341
T by hand=926.955648
V by hand=-195.497817
m2=41.9
R=  0.0364300575
Energywithoutmass = 1402.61653
Energy=  29570.6165
R=  0.0364300575
normx2 (computationally)=  -0.663649114
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099945
normx1 (by hand)=  -0.397099946
```

```
0 to x2 (h4 computationally)=  13.5674661
0 to x2 (h4 by hand)=  13.5674655
x2 to x3 (h5 computationally)=  2.89124609
x2 to x3 (h5 by hand)=  2.89124836
x2 to x3 (h5 by handdddd)=  2.8912474
x3 to x1 (h6 computationally)=  3.4391113
x3 to x1 (h6 by hand)=  3.43911229
K22  (computationally) = 0.85446127
K22  (by hand) = 0.854460947
K32  (computationally) = 0.80451447
K32  (by hand) = 0.804514234
K21  (computationally) = 0.519533963
K21  (by hand) =0.519533932
K31  (computationally) = 0.514537624
K31  (by hand) = 0.514538204
K33  (computationally) = 0.767188203
K33  (by hand) = 0.767190411
K11  (computationally) = 0.449457267
K11  (by hand) = 0.449459291
x2= 0.156762645
x3= 0.174787244
x1=  0.290956436
T by hand=929.255967
V by hand=-195.737725
m2=42.
R=  0.0364300575
Energywithoutmass = 1402.17181
Energy=  29570.1718
R=  0.0364300575
normx2 (computationally)=  -0.66364917
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099932
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.600289
```

```
0 to x2 (h4 by hand)=  13.6002891
x2 to x3 (h5 computationally)=  2.89834033
x2 to x3 (h5 by hand)=  2.89834162
x2 to x3 (h5 by handdddd)=  2.89834165
x3 to x1 (h6 computationally)=  3.44772984
x3 to x1 (h6 by hand)=  3.44773065
K22  (computationally) = 0.855490161
K22  (by hand) = 0.855490198
K32  (computationally) = 0.805484464
K32  (by hand) = 0.80548449
K21  (computationally) = 0.520170264
K21  (by hand) =0.520170267
K31  (computationally) = 0.515167678
K31  (by hand) = 0.515168024
K33  (computationally) = 0.768112782
K33  (by hand) = 0.768114156
K11  (computationally) = 0.45001098
K11  (by hand) = 0.450012591
x2= 0.15657333
x3= 0.174575314
x1=  0.290594851
T by hand=931.556208
V by hand=-195.977006
m2=42.1
R=  0.0364300575
Energywithoutmass = 1401.74297
Energy=  29569.743
R=  0.0364300575
normx2 (computationally)=  -0.663649169
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099935
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.6331113
0 to x2 (h4 by hand)=  13.6331114
```

```
x2 to x3 (h5 computationally)=  2.90543637
x2 to x3 (h5 by hand)=  2.90543768
x2 to x3 (h5 by handdddd)=  2.90543769
x3 to x1 (h6 computationally)=  3.45634924
x3 to x1 (h6 by hand)=  3.4563501
K22  (computationally) = 0.856517199
K22  (by hand) = 0.856517231
K32  (computationally) = 0.806452942
K32  (by hand) = 0.806452965
K21  (computationally) = 0.520805801
K21  (by hand) =0.520805804
K31  (computationally) = 0.515797366
K31  (by hand) = 0.515797717
K33  (computationally) = 0.76903753
K33  (by hand) = 0.769038923
K11  (computationally) = 0.45056481
K11  (by hand) = 0.450566512
x2= 0.156384691
x3= 0.174364154
x1=  0.290234604
T by hand=933.856682
V by hand=-196.216253
m2=42.2
R=  0.0364300575
Energywithoutmass = 1401.32977
Energy=  29569.3298
R=  0.0364300575
normx2 (computationally)=  -0.663649158
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.665934
0 to x2 (h4 by hand)=  13.6659339
x2 to x3 (h5 computationally)=  2.91253293
```

```
x2 to x3 (h5 by hand)=  2.91253443
x2 to x3 (h5 by handdddd)=  2.91253425
x3 to x1 (h6 computationally)=  3.46496909
x3 to x1 (h6 by hand)=  3.46496996
K22  (computationally) = 0.857542912
K22  (by hand) = 0.857542876
K32  (computationally) = 0.80742021
K32  (by hand) = 0.807420183
K21  (computationally) = 0.521440573
K21  (by hand) =0.521440569
K31  (computationally) = 0.516426362
K31  (by hand) = 0.516426759
K33  (computationally) = 0.769961402
K33  (by hand) = 0.769962961
K11  (computationally) = 0.451117909
K11  (by hand) = 0.451119658
x2= 0.156196728
x3= 0.174153754
x1=  0.289875676
T by hand=936.15725
V by hand=-196.455208
m2=42.3
R=  0.0364300575
Energywithoutmass = 1400.9321
Energy=  29568.9321
R=  0.0364300575
normx2 (computationally)=  -0.663649146
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.6987573
0 to x2 (h4 by hand)=  13.6987571
x2 to x3 (h5 computationally)=  2.91962977
x2 to x3 (h5 by hand)=  2.9196315
```

x2 to x3 (h5 by handdddd)= 2.9196311

x3 to x1 (h6 computationally)= 3.4735894

x3 to x1 (h6 by hand)= 3.47359026

K22 (computationally) = 0.858567395

K22 (by hand) = 0.85856728

K32 (computationally) = 0.808386324

K32 (by hand) = 0.80838624

K21 (computationally) = 0.522074588

K21 (by hand) =0.522074577

K31 (computationally) = 0.51705462

K31 (by hand) = 0.517055071

K33 (computationally) = 0.770884217

K33 (by hand) = 0.770885966

K11 (computationally) = 0.451670291

K11 (by hand) = 0.451672048

x2= 0.156009438

x3= 0.173944109

x1= 0.289518057

T by hand=938.45791

V by hand=−196.693866

m2=42.4

R= 0.0364300575

Energywithoutmass = 1400.54985

Energy= 29568.5498

R= 0.0364300575

normx2 (computationally)= −0.663649169

normx2 (by hand)= −0.663649215

normx3 (computationally)= −0.62369752

normx3 (by hand)= −0.623697559

normx1 (computationally)= −0.397099932

normx1 (by hand)= −0.397099946

0 to x2 (h4 computationally)= 13.7315819

0 to x2 (h4 by hand)= 13.731582

x2 to x3 (h5 computationally)= 2.92672587

x2 to x3 (h5 by hand)= 2.92672718

x2 to x3 (h5 by handdddd)= 2.92672721

```
x3 to x1 (h6 computationally)=  3.48221012
x3 to x1 (h6 by hand)=  3.48221094
K22  (computationally) = 0.859591072
K22  (by hand) = 0.859591111
K32  (computationally) = 0.809351532
K32  (by hand) = 0.80935156
K21  (computationally) = 0.522707874
K21  (by hand) =0.522707878
K31  (computationally) = 0.517681907
K31  (by hand) = 0.517682258
K33  (computationally) = 0.771805135
K33  (by hand) = 0.771806531
K11  (computationally) = 0.452221824
K11  (by hand) = 0.452223458
x2= 0.155822819
x3= 0.173735212
x1=  0.289161739
T by hand=940.758631
V by hand=-196.932157
m2=42.5
R=  0.0364300575
Energywithoutmass = 1400.18291
Energy=  29568.1829
R=  0.0364300575
normx2 (computationally)=  -0.663649091
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099937
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.764405
0 to x2 (h4 by hand)=  13.7644041
x2 to x3 (h5 computationally)=  2.93382504
x2 to x3 (h5 by hand)=  2.93382776
x2 to x3 (h5 by handdddd)=  2.93382638
x3 to x1 (h6 computationally)=  3.49083152
```

```
x3 to x1 (h6 by hand)=  3.49083241
K22  (computationally) = 0.860612394
K22  (by hand) = 0.86061192
K32  (computationally) = 0.810314939
K32  (by hand) = 0.810314594
K21  (computationally) = 0.523340356
K21  (by hand) =0.523340311
K31  (computationally) = 0.518309109
K31  (by hand) = 0.518309797
K33  (computationally) = 0.772727287
K33  (by hand) = 0.772729886
K11  (computationally) = 0.452773222
K11  (by hand) = 0.452775077
x2= 0.155636857
x3= 0.17352707
x1=  0.288806723
T by hand=943.059561
V by hand=-197.1704
m2=42.6
R=  0.0364300575
Energywithoutmass = 1399.83106
Energy=  29567.8311
R=  0.0364300575
normx2 (computationally)=  -0.663649112
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.62369752
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099941
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.7972308
0 to x2 (h4 by hand)=  13.7972301
x2 to x3 (h5 computationally)=  2.94092169
x2 to x3 (h5 by hand)=  2.94092403
x2 to x3 (h5 by handdddd)=  2.94092304
x3 to x1 (h6 computationally)=  3.4994534
x3 to x1 (h6 by hand)=  3.49945435
```

```
K22  (computationally) = 0.861633642
K22  (by hand) = 0.861633307
K32  (computationally) = 0.811277865
K32  (by hand) = 0.811277621
K21  (computationally) = 0.523972168
K21  (by hand) =0.523972136
K31  (computationally) = 0.518934951
K31  (by hand) = 0.51893555
K33  (computationally) = 0.773646093
K33  (by hand) = 0.773648374
K11  (computationally) = 0.453323798
K11  (by hand) = 0.45332575
x2= 0.155451564
x3= 0.17331966
x1=  0.288452991
T by hand=945.360531
V by hand=-197.408218
m2=42.7
R=  0.0364300575
Energywithoutmass = 1399.4943
Energy=  29567.4943
R=  0.0364300575
normx2 (computationally)=  -0.663649134
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099933
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.8300572
0 to x2 (h4 by hand)=  13.8300568
x2 to x3 (h5 computationally)=  2.94801855
x2 to x3 (h5 by hand)=  2.94802051
x2 to x3 (h5 by handdddd)=  2.94801991
x3 to x1 (h6 computationally)=  3.50807548
x3 to x1 (h6 by hand)=  3.50807631
K22  (computationally) = 0.862653703
```

```
K22  (by hand) = 0.862653513
K32  (computationally) = 0.812239669
K32  (by hand) = 0.81223953
K21  (computationally) = 0.524603211
K21  (by hand) =0.524603193
K31  (computationally) = 0.519560026
K31  (by hand) = 0.519560532
K33  (computationally) = 0.774563805
K33  (by hand) = 0.774565755
K11  (computationally) = 0.453873186
K11  (by hand) = 0.453874887
x2= 0.155266927
x3= 0.173112988
x1=  0.288100533
T by hand=947.661522
V by hand=-197.645621
m2=42.8
R=  0.0364300575
Energywithoutmass = 1399.17246
Energy=  29567.1725
R=  0.0364300575
normx2 (computationally)=  -0.663649143
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099935
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.8628839
0 to x2 (h4 by hand)=  13.8628836
x2 to x3 (h5 computationally)=  2.95511601
x2 to x3 (h5 by hand)=  2.9551178
x2 to x3 (h5 by handdddd)=  2.95511737
x3 to x1 (h6 computationally)=  3.5166983
x3 to x1 (h6 by hand)=  3.51669917
K22  (computationally) = 0.86367243
K22  (by hand) = 0.863672302
```

```
K32  (computationally) = 0.813200267
K32  (by hand) = 0.813200173
K21  (computationally) = 0.525233531
K21  (by hand) =0.525233519
K31  (computationally) = 0.520184473
K31  (by hand) = 0.520184941
K33  (computationally) = 0.775480733
K33  (by hand) = 0.775482547
K11  (computationally) = 0.45442243
K11  (by hand) = 0.454424203
x2= 0.155082943
x3= 0.172907051
x1=  0.287749352
T by hand=949.962689
V by hand=-197.882884
m2=42.9
R=  0.0364300575
Energywithoutmass = 1398.86539
Energy=  29566.8654
R=  0.0364300575
normx2 (computationally)=  -0.663649103
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099932
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.8957101
0 to x2 (h4 by hand)=  13.8957094
x2 to x3 (h5 computationally)=  2.9622151
x2 to x3 (h5 by hand)=  2.96221763
x2 to x3 (h5 by handdddd)=  2.96221647
x3 to x1 (h6 computationally)=  3.52532146
x3 to x1 (h6 by hand)=  3.52532229
K22  (computationally) = 0.864689403
K22  (by hand) = 0.864689008
K32  (computationally) = 0.814159417
```

```
K32  (by hand) = 0.814159129
K21  (computationally) = 0.525863066
K21  (by hand) =0.525863028
K31  (computationally) = 0.520808492
K31  (by hand) = 0.520809136
K33  (computationally) = 0.776397732
K33  (by hand) = 0.776400176
K11  (computationally) = 0.454970862
K11  (by hand) = 0.454972603
x2= 0.154899604
x3= 0.172701846
x1=  0.287399435
T by hand=952.263947
V by hand=-198.119879
m2=43.
R=  0.0364300575
Energywithoutmass = 1398.57295
Energy=  29566.5729
R=  0.0364300575
normx2 (computationally)=  -0.663649159
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099933
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.928539
0 to x2 (h4 by hand)=  13.9285389
x2 to x3 (h5 computationally)=  2.96931173
x2 to x3 (h5 by hand)=  2.96931326
x2 to x3 (h5 by handdddd)=  2.9693131
x3 to x1 (h6 computationally)=  3.5339452
x3 to x1 (h6 by hand)=  3.53394604
K22  (computationally) = 0.865706291
K22  (by hand) = 0.865706265
K32  (computationally) = 0.815118087
K32  (by hand) = 0.815118068
```

```
K21  (computationally) = 0.526491947
K21  (by hand) =0.526491944
K31  (computationally) = 0.521431186
K31  (by hand) = 0.52143159
K33  (computationally) = 0.777311458
K33  (by hand) = 0.777313047
K11  (computationally) = 0.45551866
K11  (by hand) = 0.45552036
x2= 0.154716918
x3= 0.172497357
x1=  0.287050774
T by hand=954.56527
V by hand=-198.3565
m2=43.1
R=  0.0364300575
Energywithoutmass = 1398.29511
Energy=  29566.2951
R=  0.0364300575
normx2 (computationally)=  -0.663649076
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099946
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.9613656
0 to x2 (h4 by hand)=  13.9613644
x2 to x3 (h5 computationally)=  2.97641249
x2 to x3 (h5 by hand)=  2.97641553
x2 to x3 (h5 by handdddd)=  2.97641387
x3 to x1 (h6 computationally)=  3.54256971
x3 to x1 (h6 by hand)=  3.54257076
K22  (computationally) = 0.866720417
K22  (by hand) = 0.866719842
K32  (computationally) = 0.816074729
K32  (by hand) = 0.81607431
K21  (computationally) = 0.527120022
```

```
K21  (by hand) =0.527119967
K31  (computationally) = 0.522054059
K31  (by hand) = 0.522054825
K33  (computationally) = 0.778227309
K33  (by hand) = 0.778230186
K11  (computationally) = 0.456066606
K11  (by hand) = 0.456068789
x2= 0.154534862
x3= 0.1722936
x1=  0.286703373
T by hand=956.866854
V by hand=-198.593184
m2=43.2
R=  0.0364300575
Energywithoutmass = 1398.03159
Energy=  29566.0316
R=  0.0364300575
normx2 (computationally)=  -0.663649169
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099937
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  13.9941963
0 to x2 (h4 by hand)=  13.9941964
x2 to x3 (h5 computationally)=  2.98350855
x2 to x3 (h5 by hand)=  2.98350991
x2 to x3 (h5 by handdddd)=  2.98350993
x3 to x1 (h6 computationally)=  3.55119416
x3 to x1 (h6 by hand)=  3.55119508
K22  (computationally) = 0.867735378
K22  (by hand) = 0.867735418
K32  (computationally) = 0.817031426
K32  (by hand) = 0.817031456
K21  (computationally) = 0.52774745
K21  (by hand) =0.527747454
```

```
K31  (computationally) = 0.52267505
K31  (by hand) = 0.522675413
K33  (computationally) = 0.779138057
K33  (by hand) = 0.779139502
K11  (computationally) = 0.456612639
K11  (by hand) = 0.456614459
x2= 0.154353459
x3= 0.172090544
x1=  0.286357196
T by hand=959.168289
V by hand=-198.829094
m2=43.3
R=  0.0364300575
Energywithoutmass = 1397.78251
Energy=  29565.7825
R=  0.0364300575
normx2 (computationally)=  -0.663649075
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099932
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.0270238
0 to x2 (h4 by hand)=  14.0270227
x2 to x3 (h5 computationally)=  2.99061009
x2 to x3 (h5 by hand)=  2.99061315
x2 to x3 (h5 by handdddd)=  2.99061148
x3 to x1 (h6 computationally)=  3.5598192
x3 to x1 (h6 by hand)=  3.55982004
K22  (computationally) = 0.868747035
K22  (by hand) = 0.868746456
K32  (computationally) = 0.817985786
K32  (by hand) = 0.817985363
K21  (computationally) = 0.528374023
K21  (by hand) =0.528373967
K31  (computationally) = 0.52329651
```

K31 (by hand) = 0.523297282

K33 (computationally) = 0.780052045

K33 (by hand) = 0.780054946

K11 (computationally) = 0.457158526

K11 (by hand) = 0.457160321

x2= 0.154172675

x3= 0.171888216

x1= 0.286012262

T by hand=961.469956

V by hand=-199.065047

m2=43.4

R= 0.0364300575

Energywithoutmass = 1397.54749

Energy= 29565.5475

R= 0.0364300575

normx2 (computationally)= -0.663649075

normx2 (by hand)= -0.663649215

normx3 (computationally)= -0.623697519

normx3 (by hand)= -0.623697559

normx1 (computationally)= -0.397099932

normx1 (by hand)= -0.397099946

0 to x2 (h4 computationally)= 14.0598539

0 to x2 (h4 by hand)= 14.0598527

x2 to x3 (h5 computationally)= 2.99770924

x2 to x3 (h5 by hand)= 2.9977123

x2 to x3 (h5 by handdddd)= 2.99771063

x3 to x1 (h6 computationally)= 3.56844482

x3 to x1 (h6 by hand)= 3.56844567

K22 (computationally) = 0.869758594

K22 (by hand) = 0.869758014

K32 (computationally) = 0.818939663

K32 (by hand) = 0.818939239

K21 (computationally) = 0.528999953

K21 (by hand) =0.528999897

K31 (computationally) = 0.523916674

K31 (by hand) = 0.523917449

```
K33  (computationally) = 0.780962831
K33  (by hand) = 0.780965742
K11  (computationally) = 0.457703852
K11  (by hand) = 0.457705654
x2= 0.153992528
x3= 0.171686588
x1=  0.285668556
T by hand=963.771699
V by hand=-199.300647
m2=43.5
R=  0.0364300575
Energywithoutmass = 1397.32659
Energy=  29565.3266
R=  0.0364300575
normx2 (computationally)=  -0.663649138
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099934
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.0926858
0 to x2 (h4 by hand)=  14.0926854
x2 to x3 (h5 computationally)=  3.00480685
x2 to x3 (h5 by hand)=  3.00480879
x2 to x3 (h5 by handdddd)=  3.00480825
x3 to x1 (h6 computationally)=  3.57707096
x3 to x1 (h6 by hand)=  3.57707184
K22  (computationally) = 0.870769711
K22  (by hand) = 0.870769546
K32  (computationally) = 0.81989286
K32  (by hand) = 0.81989274
K21  (computationally) = 0.529625214
K21  (by hand) =0.529625198
K31  (computationally) = 0.524535731
K31  (by hand) = 0.524536234
K33  (computationally) = 0.781871122
```

K33 (by hand) = 0.781873067

K11 (computationally) = 0.458248535

K11 (by hand) = 0.458250326

x2= 0.153813014

x3= 0.171485658

x1= 0.285326068

T by hand=966.073513

V by hand=−199.535906

m2=43.6

R= 0.0364300575

Energywithoutmass = 1397.11968

Energy= 29565.1197

R= 0.0364300575

normx2 (computationally)= −0.663649168

normx2 (by hand)= −0.663649215

normx3 (computationally)= −0.623697519

normx3 (by hand)= −0.623697559

normx1 (computationally)= −0.397099937

normx1 (by hand)= −0.397099946

0 to x2 (h4 computationally)= 14.1255176

0 to x2 (h4 by hand)= 14.1255177

x2 to x3 (h5 computationally)= 3.0119056

x2 to x3 (h5 by hand)= 3.01190698

x2 to x3 (h5 by handdddd)= 3.011907

x3 to x1 (h6 computationally)= 3.58569759

x3 to x1 (h6 by hand)= 3.58569851

K22 (computationally) = 0.871779302

K22 (by hand) = 0.871779342

K32 (computationally) = 0.820844753

K32 (by hand) = 0.820844782

K21 (computationally) = 0.530249734

K21 (by hand) =0.530249738

K31 (computationally) = 0.525154279

K31 (by hand) = 0.52515465

K33 (computationally) = 0.782779106

K33 (by hand) = 0.782780581

```
K11  (computationally) = 0.458792665
K11  (by hand) = 0.458794497
x2= 0.15363412
x3= 0.171285429
x1=  0.284984795
T by hand=968.375446
V by hand=-199.770947
m2=43.7
R=  0.0364300575
Energywithoutmass = 1396.9266
Energy=  29564.9266
R=  0.0364300575
normx2 (computationally)=  -0.663649125
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099943
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.1583484
0 to x2 (h4 by hand)=  14.1583479
x2 to x3 (h5 computationally)=  3.01900666
x2 to x3 (h5 by hand)=  3.01900884
x2 to x3 (h5 by handdddd)=  3.01900807
x3 to x1 (h6 computationally)=  3.59432477
x3 to x1 (h6 by hand)=  3.59432579
K22  (computationally) = 0.872786892
K22  (by hand) = 0.872786644
K32  (computationally) = 0.821795067
K32  (by hand) = 0.821794886
K21  (computationally) = 0.530873491
K21  (by hand) =0.530873467
K31  (computationally) = 0.525772596
K31  (by hand) = 0.525773157
K33  (computationally) = 0.783687758
K33  (by hand) = 0.783689911
K11  (computationally) = 0.459336456
```

```
K11  (by hand) = 0.459338522
x2= 0.15345584
x3= 0.171085903
x1=  0.284644732
T by hand=970.677542
V by hand=-200.005866
m2=43.8
R=  0.0364300575
Energywithoutmass = 1396.74719
Energy=  29564.7472
R=  0.0364300575
normx2 (computationally)=  -0.663649109
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099946
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.1911804
0 to x2 (h4 by hand)=  14.1911797
x2 to x3 (h5 computationally)=  3.02610719
x2 to x3 (h5 by hand)=  3.02610967
x2 to x3 (h5 by handdddd)=  3.02610861
x3 to x1 (h6 computationally)=  3.60295236
x3 to x1 (h6 by hand)=  3.60295343
K22  (computationally) = 0.873793642
K22  (by hand) = 0.873793287
K32  (computationally) = 0.822744475
K32  (by hand) = 0.822744216
K21  (computationally) = 0.531496549
K21  (by hand) =0.531496515
K31  (computationally) = 0.526390028
K31  (by hand) = 0.526390661
K33  (computationally) = 0.784594748
K33  (by hand) = 0.784597159
K11  (computationally) = 0.459879464
K11  (by hand) = 0.459881657
```

```
x2= 0.153278179
x3= 0.170887066
x1=  0.284305866
T by hand=972.979701
V by hand=-200.240449
m2=43.9
R=  0.0364300575
Energywithoutmass = 1396.58142
Energy=  29564.5814
R=  0.0364300575
normx2 (computationally)=  -0.663649116
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099946
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.2240135
0 to x2 (h4 by hand)=  14.2240128
x2 to x3 (h5 computationally)=  3.0332073
x2 to x3 (h5 by hand)=  3.03320966
x2 to x3 (h5 by handdddd)=  3.03320872
x3 to x1 (h6 computationally)=  3.61158033
x3 to x1 (h6 by hand)=  3.6115814
K22  (computationally) = 0.87479951
K22  (by hand) = 0.874799202
K32  (computationally) = 0.823692956
K32  (by hand) = 0.823692731
K21  (computationally) = 0.532118906
K21  (by hand) =0.532118876
K31  (computationally) = 0.527006601
K31  (by hand) = 0.527007206
K33  (computationally) = 0.785500171
K33  (by hand) = 0.78550248
K11  (computationally) = 0.460421653
K11  (by hand) = 0.460423842
x2= 0.153101131
```

x3= 0.170688914

x1= 0.283968189

T by hand=975.281918

V by hand=−200.47469

m2=44.

R= 0.0364300575

Energywithoutmass = 1396.42917

Energy= 29 564.4292

R= 0.0364300575

normx2 (computationally)= −0.663649121

normx2 (by hand)= −0.663649215

normx3 (computationally)= −0.623697519

normx3 (by hand)= −0.623697559

normx1 (computationally)= −0.397099943

normx1 (by hand)= −0.397099946

0 to x2 (h4 computationally)= 14.256847

0 to x2 (h4 by hand)= 14.2568465

x2 to x3 (h5 computationally)= 3.04030771

x2 to x3 (h5 by hand)= 3.04030999

x2 to x3 (h5 by handdddd)= 3.04030914

x3 to x1 (h6 computationally)= 3.62020868

x3 to x1 (h6 by hand)= 3.62020971

K22 (computationally) = 0.875804209

K22 (by hand) = 0.875803931

K32 (computationally) = 0.824640343

K32 (by hand) = 0.82464014

K21 (computationally) = 0.532740546

K21 (by hand) =0.532740519

K31 (computationally) = 0.52762248

K31 (by hand) = 0.527623066

K33 (computationally) = 0.78640462

K33 (by hand) = 0.786406864

K11 (computationally) = 0.460963087

K11 (by hand) = 0.460965183

x2= 0.152924693

x3= 0.170491446

```
x1=  0.283631695
T by hand=977.584211
V by hand=-200.708634
m2=44.1
R=  0.0364300575
Energywithoutmass = 1396.29033
Energy=  29564.2903
R=  0.0364300575
normx2 (computationally)=  -0.663649122
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099946
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.2896811
0 to x2 (h4 by hand)=  14.2896805
x2 to x3 (h5 computationally)=  3.04740845
x2 to x3 (h5 by hand)=  3.04741072
x2 to x3 (h5 by handdddd)=  3.04740989
x3 to x1 (h6 computationally)=  3.62883766
x3 to x1 (h6 by hand)=  3.62883875
K22  (computationally) = 0.876807727
K22  (by hand) = 0.876807456
K32  (computationally) = 0.825586632
K32  (by hand) = 0.825586434
K21  (computationally) = 0.533361492
K21  (by hand) =0.533361466
K31  (computationally) = 0.528237699
K31  (by hand) = 0.528238283
K33  (computationally) = 0.787308131
K33  (by hand) = 0.787310369
K11  (computationally) = 0.461504216
K11  (by hand) = 0.46150642
x2= 0.15274886
x3= 0.170294658
x1=  0.283296383
```

```
T by hand=979.886646
V by hand=-200.942392
m2=44.2
R=  0.0364300575
Energywithoutmass = 1396.16477
Energy=  29564.1648
R=  0.0364300575
normx2 (computationally)=  -0.663649159
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099933
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.3225164
0 to x2 (h4 by hand)=  14.3225164
x2 to x3 (h5 computationally)=  3.05450838
x2 to x3 (h5 by hand)=  3.05450996
x2 to x3 (h5 by handdddd)=  3.05450982
x3 to x1 (h6 computationally)=  3.6374667
x3 to x1 (h6 by hand)=  3.63746758
K22  (computationally) = 0.877810535
K22  (by hand) = 0.877810517
K32  (computationally) = 0.826532097
K32  (by hand) = 0.826532084
K21  (computationally) = 0.533981725
K21  (by hand) =0.533981723
K31  (computationally) = 0.528851948
K31  (by hand) = 0.528852368
K33  (computationally) = 0.788209763
K33  (by hand) = 0.788211418
K11  (computationally) = 0.462043909
K11  (by hand) = 0.462045687
x2= 0.152573632
x3= 0.170098542
x1=  0.282962232
T by hand=982.189042
```

```
V by hand=-201.175643
m2=44.3
R=  0.0364300575
Energywithoutmass = 1396.05244
Energy=  29564.0524
R=  0.0364300575
normx2 (computationally)=  -0.663649139
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099937
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.3553512
0 to x2 (h4 by hand)=  14.3553509
x2 to x3 (h5 computationally)=  3.06161018
x2 to x3 (h5 by hand)=  3.06161214
x2 to x3 (h5 by handdddd)=  3.06161162
x3 to x1 (h6 computationally)=  3.6460966
x3 to x1 (h6 by hand)=  3.64609755
K22  (computationally) = 0.878811544
K22  (by hand) = 0.878811392
K32  (computationally) = 0.827476109
K32  (by hand) = 0.827475997
K21  (computationally) = 0.53460125
K21  (by hand) =0.534601236
K31  (computationally) = 0.529465909
K31  (by hand) = 0.52946642
K33  (computationally) = 0.789111722
K33  (by hand) = 0.7891137
K11  (computationally) = 0.462583851
K11  (by hand) = 0.462585764
x2= 0.152398998
x3= 0.169903103
x1=  0.282629256
T by hand=984.491677
V by hand=-201.408905
```

```
m2=44.4
R=  0.0364300575
Energywithoutmass = 1395.95312
Energy=  29563.9531
R=  0.0364300575
normx2 (computationally)=  -0.663649083
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099932
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.3881857
0 to x2 (h4 by hand)=  14.3881846
x2 to x3 (h5 computationally)=  3.06871328
x2 to x3 (h5 by hand)=  3.06871631
x2 to x3 (h5 by handdddd)=  3.06871473
x3 to x1 (h6 computationally)=  3.65472676
x3 to x1 (h6 by hand)=  3.65472764
K22  (computationally) = 0.87981099
K22  (by hand) = 0.87981045
K32  (computationally) = 0.828418803
K32  (by hand) = 0.828418409
K21  (computationally) = 0.53522003
K21  (by hand) =0.535219978
K31  (computationally) = 0.530079402
K31  (by hand) = 0.530080169
K33  (computationally) = 0.790013546
K33  (by hand) = 0.790016439
K11  (computationally) = 0.463122868
K11  (by hand) = 0.463124718
x2= 0.152224956
x3= 0.169708336
x1=  0.282297435
T by hand=986.794374
V by hand=-201.641865
m2=44.5
```

```
R=  0.0364300575
Energywithoutmass = 1395.86677
Energy=  29563.8668
R=  0.0364300575
normx2 (computationally)=  -0.663649167
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.4210237
0 to x2 (h4 by hand)=  14.4210238
x2 to x3 (h5 computationally)=  3.07581252
x2 to x3 (h5 by hand)=  3.07581398
x2 to x3 (h5 by handdddd)=  3.07581397
x3 to x1 (h6 computationally)=  3.6633576
x3 to x1 (h6 by hand)=  3.66335854
K22  (computationally) = 0.88081097
K22  (by hand) = 0.880811006
K32  (computationally) = 0.829361402
K32  (by hand) = 0.829361428
K21  (computationally) = 0.535838242
K21  (by hand) =0.535838245
K31  (computationally) = 0.530691304
K31  (by hand) = 0.530691695
K33  (computationally) = 0.790911012
K33  (by hand) = 0.790912567
K11  (computationally) = 0.463661445
K11  (by hand) = 0.46366332
x2= 0.152051518
x3= 0.16951422
x1=  0.281966765
T by hand=989.097137
V by hand=-201.874453
m2=44.6
R=  0.0364300575
```

```
Energywithoutmass = 1395.79337
Energy=  29563.7934
R=  0.0364300575
normx2 (computationally)=  -0.663649112
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.4538594
0 to x2 (h4 by hand)=  14.4538587
x2 to x3 (h5 computationally)=  3.08291602
x2 to x3 (h5 by hand)=  3.08291852
x2 to x3 (h5 by handdddd)=  3.08291748
x3 to x1 (h6 computationally)=  3.67198879
x3 to x1 (h6 by hand)=  3.67198974
K22  (computationally) = 0.881808193
K22  (by hand) = 0.881807852
K32  (computationally) = 0.830301992
K32  (by hand) = 0.830301744
K21  (computationally) = 0.536455643
K21  (by hand) =0.536455611
K31  (computationally) = 0.531303419
K31  (by hand) = 0.531304059
K33  (computationally) = 0.791810774
K33  (by hand) = 0.791813218
K11  (computationally) = 0.464199477
K11  (by hand) = 0.464201435
x2= 0.151878657
x3= 0.169320777
x1=  0.281637242
T by hand=991.400053
V by hand=-202.106944
m2=44.7
R=  0.0364300575
Energywithoutmass = 1395.73265
```

```
Energy=  29563.7327
R=  0.0364300575
normx2 (computationally)=  -0.663649109
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.4866967
0 to x2 (h4 by hand)=  14.486696
x2 to x3 (h5 computationally)=  3.09001823
x2 to x3 (h5 by hand)=  3.09002079
x2 to x3 (h5 by handdddd)=  3.0900197
x3 to x1 (h6 computationally)=  3.68062043
x3 to x1 (h6 by hand)=  3.68062138
K22  (computationally) = 0.882804914
K22  (by hand) = 0.882804555
K32  (computationally) = 0.831241888
K32  (by hand) = 0.831241626
K21  (computationally) = 0.537072393
K21  (by hand) =0.537072359
K31  (computationally) = 0.531914506
K31  (by hand) = 0.531915161
K33  (computationally) = 0.792708291
K33  (by hand) = 0.792710787
K11  (computationally) = 0.464736804
K11  (by hand) = 0.464738771
x2= 0.151706385
x3= 0.169127986
x1=  0.281308855
T by hand=993.703027
V by hand=-202.339093
m2=44.8
R=  0.0364300575
Energywithoutmass = 1395.68464
Energy=  29563.6846
```

```
R=  0.0364300575
normx2 (computationally)=  -0.663649105
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5195346
0 to x2 (h4 by hand)=  14.5195338
x2 to x3 (h5 computationally)=  3.09712072
x2 to x3 (h5 by hand)=  3.09712337
x2 to x3 (h5 by handdddd)=  3.09712219
x3 to x1 (h6 computationally)=  3.68925252
x3 to x1 (h6 by hand)=  3.68925347
K22  (computationally) = 0.883800502
K22  (by hand) = 0.883800111
K32  (computationally) = 0.832180722
K32  (by hand) = 0.832180436
K21  (computationally) = 0.537688451
K21  (by hand) =0.537688414
K31  (computationally) = 0.532524922
K31  (by hand) = 0.5325256
K33  (computationally) = 0.793604849
K33  (by hand) = 0.793607426
K11  (computationally) = 0.465273524
K11  (by hand) = 0.465275498
x2= 0.151534696
x3= 0.168935849
x1=  0.280981599
T by hand=996.006094
V by hand=-202.570984
m2=44.9
R=  0.0364300575
Energywithoutmass = 1395.64919
Energy=  29563.6492
R=  0.0364300575
```

```
normx2 (computationally)=  -0.663649167
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5523744
0 to x2 (h4 by hand)=  14.5523745
x2 to x3 (h5 computationally)=  3.10422146
x2 to x3 (h5 by hand)=  3.10422293
x2 to x3 (h5 by handdddd)=  3.10422294
x3 to x1 (h6 computationally)=  3.69788506
x3 to x1 (h6 by hand)=  3.69788601
K22  (computationally) = 0.884795781
K22  (by hand) = 0.884795824
K32  (computationally) = 0.833118975
K32  (by hand) = 0.833119006
K21  (computationally) = 0.538303875
K21  (by hand) =0.538303879
K31  (computationally) = 0.533134213
K31  (by hand) = 0.533134608
K33  (computationally) = 0.794498793
K33  (by hand) = 0.794500363
K11  (computationally) = 0.465809531
K11  (by hand) = 0.46581143
x2= 0.15136359
x3= 0.168744355
x1=  0.280655466
T by hand=998.309213
V by hand=-202.802514
m2=45.
R=  0.0364300575
Energywithoutmass = 1395.62626
Energy=  29563.6263
R=  0.0364300575
normx2 (computationally)=  -0.663649163
```

```
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5852134
0 to x2 (h4 by hand)=  14.5852134
x2 to x3 (h5 computationally)=  3.11132438
x2 to x3 (h5 by hand)=  3.11132593
x2 to x3 (h5 by handdddd)=  3.11132586
x3 to x1 (h6 computationally)=  3.70651805
x3 to x1 (h6 by hand)=  3.706519
K22  (computationally) = 0.885789162
K22  (by hand) = 0.885789178
K32  (computationally) = 0.834055726
K32  (by hand) = 0.834055737
K21  (computationally) = 0.538918562
K21  (by hand) =0.538918563
K31  (computationally) = 0.533743266
K31  (by hand) = 0.53374368
K33  (computationally) = 0.795393343
K33  (by hand) = 0.795394985
K11  (computationally) = 0.466345055
K11  (by hand) = 0.46634696
x2= 0.151193055
x3= 0.168553512
x1=  0.280330454
T by hand=1000.61246
V by hand=-203.033887
m2=45.1
R=  0.0364300575
Energywithoutmass = 1395.61565
Energy=  29563.6156
R=  0.0364300575
normx2 (computationally)=  -0.663649143
normx2 (by hand)=  -0.663649215
```

```
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.6180525
0 to x2 (h4 by hand)=  14.6180523
x2 to x3 (h5 computationally)=  3.11842799
x2 to x3 (h5 by hand)=  3.11842993
x2 to x3 (h5 by handdddd)=  3.11842948
x3 to x1 (h6 computationally)=  3.71515148
x3 to x1 (h6 by hand)=  3.71515243
K22  (computationally) = 0.886781251
K22  (by hand) = 0.886781127
K32  (computationally) = 0.834991326
K32  (by hand) = 0.834991236
K21  (computationally) = 0.539532554
K21  (by hand) =0.539532542
K31  (computationally) = 0.534351748
K31  (by hand) = 0.534352255
K33  (computationally) = 0.796287287
K33  (by hand) = 0.796289261
K11  (computationally) = 0.46688001
K11  (by hand) = 0.466881941
x2= 0.15102309
x3= 0.168363313
x1=  0.280006555
T by hand=1002.91582
V by hand=-203.265028
m2=45.2
R=  0.0364300575
Energywithoutmass = 1395.6173
Energy=  29563.6173
R=  0.0364300575
normx2 (computationally)=  -0.663649114
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
```

```
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5884962
0 to x2 (h4 by hand)=  14.5884956
x2 to x3 (h5 computationally)=  3.11203612
x2 to x3 (h5 by hand)=  3.11203861
x2 to x3 (h5 by handdddd)=  3.11203761
x3 to x1 (h6 computationally)=  3.70738137
x3 to x1 (h6 by hand)=  3.70738232
K22  (computationally) = 0.885887857
K22  (by hand) = 0.88588753
K32  (computationally) = 0.834149004
K32  (by hand) = 0.834148766
K21  (computationally) = 0.538979955
K21  (by hand) =0.538979923
K31  (computationally) = 0.533804457
K31  (by hand) = 0.533805096
K33  (computationally) = 0.795483921
K33  (by hand) = 0.795486363
K11  (computationally) = 0.466398656
K11  (by hand) = 0.466400621
x2= 0.151176029
x3= 0.168534468
x1=  0.280298016
T by hand=1000.84282
V by hand=-203.057084
m2=45.11
R=  0.0364300575
Energywithoutmass = 1395.61523
Energy=  29563.6152
R=  0.0364300575
normx2 (computationally)=  -0.663649085
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
```

```
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5917795
0 to x2 (h4 by hand)=  14.5917785
x2 to x3 (h5 computationally)=  3.11274728
x2 to x3 (h5 by hand)=  3.11275033
x2 to x3 (h5 by handdddd)=  3.11274876
x3 to x1 (h6 computationally)=  3.7082447
x3 to x1 (h6 by hand)=  3.70824565
K22  (computationally) = 0.885986778
K22  (by hand) = 0.885986247
K32  (computationally) = 0.834242411
K32  (by hand) = 0.834242024
K21  (computationally) = 0.539041356
K21  (by hand) =0.539041305
K31  (computationally) = 0.53386551
K31  (by hand) = 0.533866284
K33  (computationally) = 0.795574009
K33  (by hand) = 0.795576928
K11  (computationally) = 0.466452218
K11  (by hand) = 0.466454219
x2= 0.15115901
x3= 0.168515428
x1=  0.280265588
T by hand=1001.07317
V by hand=-203.080248
m2=45.12
R=  0.0364300575
Energywithoutmass = 1395.61495
Energy=  29563.6149
R=  0.0364300575
normx2 (computationally)=  -0.663649167
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
```

```
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5950653
0 to x2 (h4 by hand)=  14.5950653
x2 to x3 (h5 computationally)=  3.11345515
x2 to x3 (h5 by hand)=  3.11345663
x2 to x3 (h5 by handdddd)=  3.11345664
x3 to x1 (h6 computationally)=  3.70910803
x3 to x1 (h6 by hand)=  3.70910898
K22  (computationally) = 0.886087022
K22  (by hand) = 0.886087064
K32  (computationally) = 0.834336583
K32  (by hand) = 0.834336613
K21  (computationally) = 0.539102839
K21  (by hand) =0.539102843
K31  (computationally) = 0.533925816
K31  (by hand) = 0.533926214
K33  (computationally) = 0.795661394
K33  (by hand) = 0.795662975
K11  (computationally) = 0.466505587
K11  (by hand) = 0.466507488
x2= 0.151142006
x3= 0.168496385
x1=  0.280233168
T by hand=1001.30345
V by hand=-203.103241
m2=45.13
R=  0.0364300575
Energywithoutmass = 1395.61487
Energy=  29563.6149
R=  0.0364300575
normx2 (computationally)=  -0.663649158
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
```

```
0 to x2 (h4 computationally)=  14.598349
0 to x2 (h4 by hand)=  14.598349
x2 to x3 (h5 computationally)=  3.11416571
x2 to x3 (h5 by hand)=  3.11416737
x2 to x3 (h5 by handdddd)=  3.1141672
x3 to x1 (h6 computationally)=  3.70997137
x3 to x1 (h6 by hand)=  3.70997232
K22  (computationally) = 0.886186164
K22  (by hand) = 0.886186144
K32  (computationally) = 0.83443011
K32  (by hand) = 0.834430096
K21  (computationally) = 0.539164243
K21  (by hand) =0.539164241
K31  (computationally) = 0.53398672
K31  (by hand) = 0.533987159
K33  (computationally) = 0.79575097
K33  (by hand) = 0.795752698
K11  (computationally) = 0.466559103
K11  (by hand) = 0.466561015
x2= 0.151125001
x3= 0.168477356
x1=  0.280200761
T by hand=1001.53379
V by hand=-203.12637
m2=45.14
R=  0.0364300575
Energywithoutmass = 1395.61485
Energy=  29563.6148
R=  0.0364300575
normx2 (computationally)=  -0.663649138
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.6016326
```

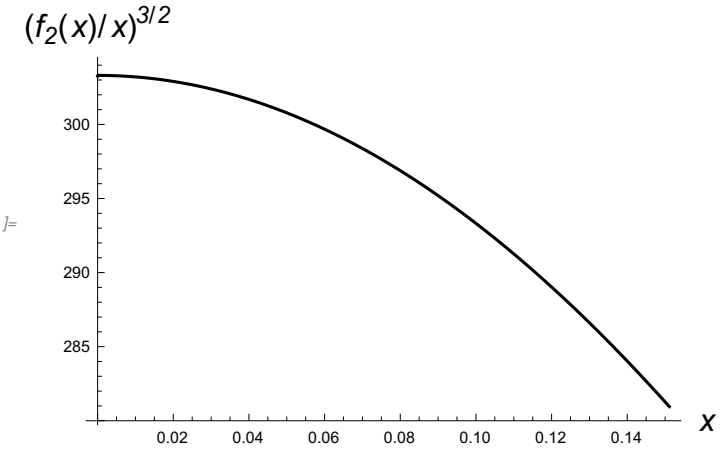
```
0 to x2 (h4 by hand)=  14.6016322
x2 to x3 (h5 computationally)=  3.11487659
x2 to x3 (h5 by hand)=  3.11487863
x2 to x3 (h5 by handdddd)=  3.11487808
x3 to x1 (h6 computationally)=  3.71083471
x3 to x1 (h6 by hand)=  3.71083566
K22  (computationally) = 0.886285167
K22  (by hand) = 0.886285009
K32  (computationally) = 0.834523552
K32  (by hand) = 0.834523437
K21  (computationally) = 0.539225631
K21  (by hand) =0.539225616
K31  (computationally) = 0.53404769
K31  (by hand) = 0.534048219
K33  (computationally) = 0.795840798
K33  (by hand) = 0.795842847
K11  (computationally) = 0.466612632
K11  (by hand) = 0.466614567
x2= 0.151108
x3= 0.168458334
x1=  0.280168366
T by hand=1001.76414
V by hand=-203.149512
m2=45.15
R=  0.0364300575
Energywithoutmass = 1395.61494
Energy=  29563.6149
R=  0.0364300575
normx2 (computationally)=  -0.663649167
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5986776
0 to x2 (h4 by hand)=  14.5986777
```

```
x2 to x3 (h5 computationally)=  3.11423648
x2 to x3 (h5 by hand)=  3.11423797
x2 to x3 (h5 by handdddd)=  3.11423797
x3 to x1 (h6 computationally)=  3.7100577
x3 to x1 (h6 by hand)=  3.71005865
K22  (computationally) = 0.886196193
K22  (by hand) = 0.886196234
K32  (computationally) = 0.83443953
K32  (by hand) = 0.834439559
K21  (computationally) = 0.53917039
K21  (by hand) =0.539170394
K31  (computationally) = 0.533992746
K31  (by hand) = 0.533993145
K33  (computationally) = 0.795759694
K33  (by hand) = 0.795761279
K11  (computationally) = 0.466564438
K11  (by hand) = 0.466566339
x2= 0.151123301
x3= 0.168475452
x1=  0.280197521
T by hand=1001.55682
V by hand=-203.128668
m2=45.141
R=  0.0364300575
Energywithoutmass = 1395.61486
Energy=  29563.6149
R=  0.0364300575
normx2 (computationally)=  -0.663649086
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5990043
0 to x2 (h4 by hand)=  14.5990032
x2 to x3 (h5 computationally)=  3.11430989
```

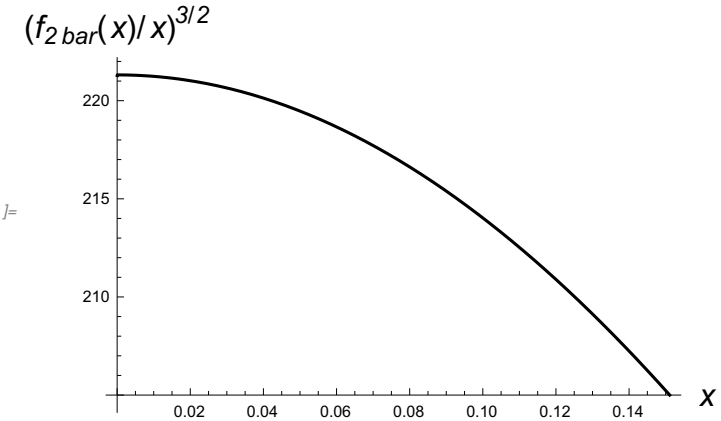
```
x2 to x3 (h5 by hand)=  3.11431291
x2 to x3 (h5 by handdddd)=  3.11431138
x3 to x1 (h6 computationally)=  3.71014403
x3 to x1 (h6 by hand)=  3.71014499
K22  (computationally) = 0.886205155
K22  (by hand) = 0.886204635
K32  (computationally) = 0.834448328
K32  (by hand) = 0.834447949
K21  (computationally) = 0.539176467
K21  (by hand) =0.539176418
K31  (computationally) = 0.533999365
K31  (by hand) = 0.534000132
K33  (computationally) = 0.795770574
K33  (by hand) = 0.795773469
K11  (computationally) = 0.466569923
K11  (by hand) = 0.466571922
x2= 0.151121595
x3= 0.168473557
x1=  0.280194283
T by hand=1001.5799
V by hand=-203.131101
m2=45.142
R=  0.0364300575
Energywithoutmass = 1395.61481
Energy=  29563.6148
R=  0.0364300575
normx2 (computationally)=  -0.663649163
normx2 (by hand)=  -0.663649215
normx3 (computationally)=  -0.623697519
normx3 (by hand)=  -0.623697559
normx1 (computationally)=  -0.397099936
normx1 (by hand)=  -0.397099946
0 to x2 (h4 computationally)=  14.5993343
0 to x2 (h4 by hand)=  14.5993343
x2 to x3 (h5 computationally)=  3.11437866
x2 to x3 (h5 by hand)=  3.11438022
```

```
x2 to x3 (h5 by handdddd)=  3.11438014
x3 to x1 (h6 computationally)=  3.71023037
x3 to x1 (h6 by hand)=  3.71023132
K22  (computationally) = 0.886215996
K22  (by hand) = 0.88621601
K32  (computationally) = 0.83445822
K32  (by hand) = 0.83445823
K21  (computationally) = 0.539182669
K21  (by hand) =0.53918267
K31  (computationally) = 0.53400494
K31  (by hand) = 0.534005356
K33  (computationally) = 0.795777657
K33  (by hand) = 0.795779304
K11  (computationally) = 0.466575144
K11  (by hand) = 0.46657705
x2= 0.151119901
x3= 0.168471648
x1=  0.280191041
T by hand=1001.60289
V by hand=-203.133296
m2=45.143
R=  0.0364300575
Energywithoutmass = 1395.61487
Energy=  29563.6149
x1=0.280191041
x3=0.168471648
x2=0.151119901
Kinetic Energy=1001.60289
Potential Energy=-203.133296
Volume Energy=597.145276
```

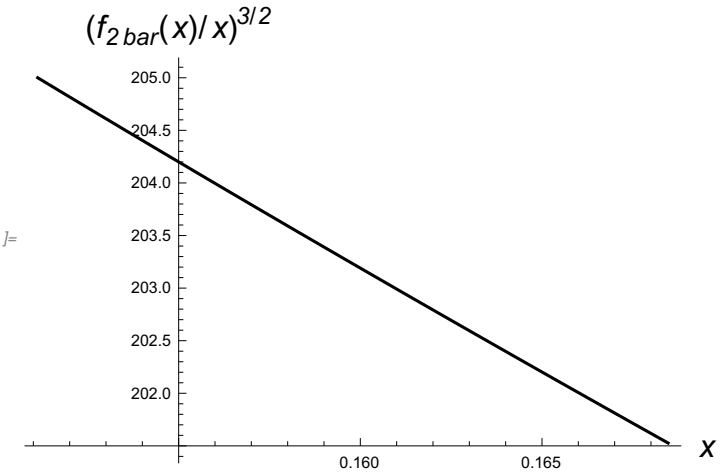
Out[ ]=


$(f_2(x)/x)^{3/2}$
300
295
290
285
0.02
0.04
0.06
0.08
0.10
0.12
0.14
x

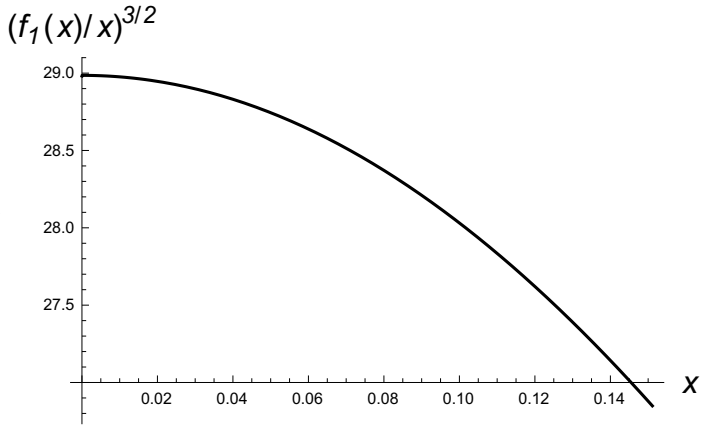
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
220
215
210
0.02
0.04
0.06
0.08
0.10
0.12
0.14
x

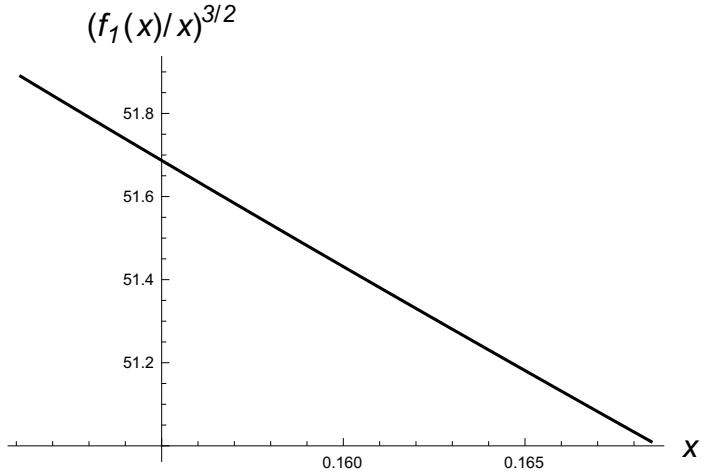
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
205.0
204.5
204.0
203.5
203.0
202.5
202.0
0.160
0.165
x

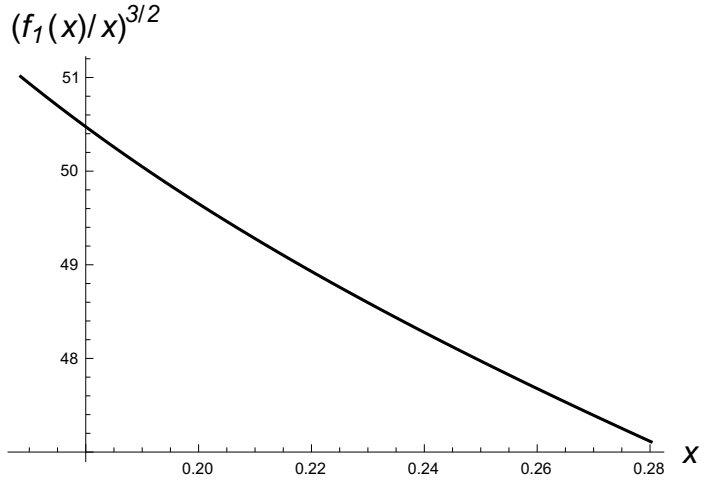
Out[ ]=


$(f_1(x)/x)^{3/2}$
29.0
28.5
28.0
27.5
0.02
0.04
0.06
0.08
0.10
0.12
0.14
x

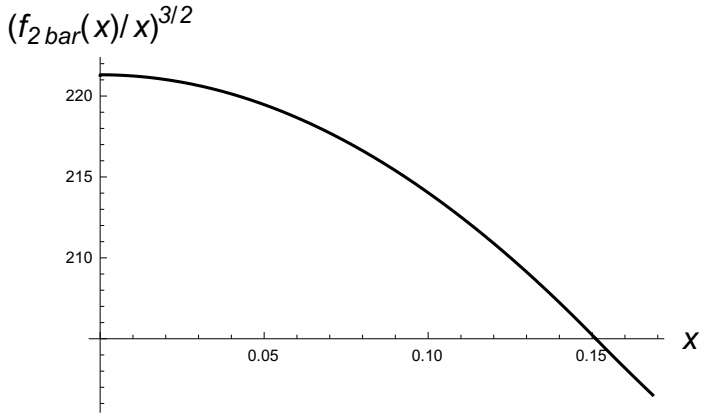
Out[ ]=


$(f_1(x)/x)^{3/2}$
51.8
51.6
51.4
51.2
0.160
0.165
x

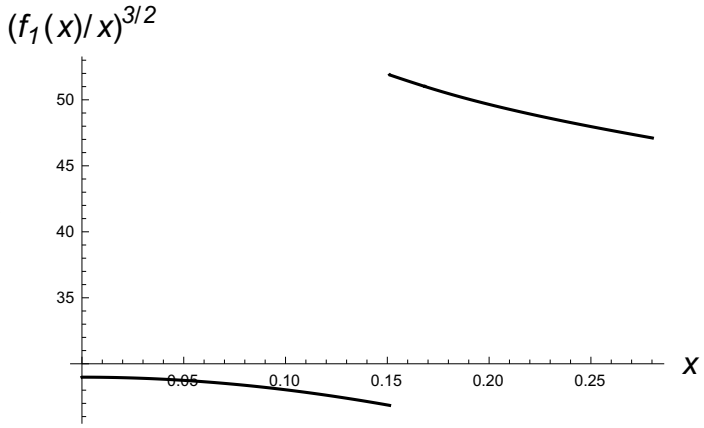
Out[ ]=


$(f_1(x)/x)^{3/2}$
51
50
49
48
0.20
0.22
0.24
0.26
0.28
x

Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
220
215
210
0.05
0.10
0.15
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
50
45
40
35
0.05
0.10
0.15
0.20
0.25
x